8) Explain alter command with various clauses using Syntax & example.

=> The SQL alter command is a Data Definition language statement used to modify the structure of an existing database table.

1) Rename Clause :-

The rename clause used to changes the name of a table or an existing coloumn without losing data.

Syntax :-

Alter table table_name rename to new_table name;

Eg :

Alter table Students Rename To learners;

2) Add Clause :-

The Add Clause lets you insert a brand new coloumn into an existing table.

Syntax:-
Alter table table_name ADD Coloumn_name datatype;

Eg:
Alter table employee ADD email Varchar(100);

3] Drop Clause :-
The drop clause removes an existing coloumn entirely from a table schema.
Syntax:
Alter table table_name DROP column_name;

Eg:
Alter table employee drop email;

4] Group by Clause :-
The Groupby clause is used in a Select Statement to arrange identical data into summary group using aggregate function like Count, Sum or Avg.
Syntax:
Select coloumn_name, Count(*) from table_name GroupBy Coloumn_name;

Eg:
Select department, Count(*) from employee GroupBy Department;

5] OrderBy Clause :-
The OrderBy Clause is used in a select statement to sort the result set in ascending or descending order.

Syntax:-

Select Coloumn1, Coloumn2 from table name Orderby Coloumn1 ASC/DES;

Eg.

Select name, Salary from employee Orderby Salary DESC;

9) Explain the types of joins used in SQL with Eg.

⇒ An SQL join is used to combine rows from two or more tables based on a related coloumn between them.

1) Inner Join :-

These joins are also called equi joins. They are known as equi joins because the where clause compares two coloumns using the = operator.

Syntax :-

Select coloumn_name(s)
From table1
INNER JOIN table2 = t2.coloumn

Eg.

Select ename, dname from employee, department where employee.dept no = department.dno;

2) Outer join :-

It returns all ~~rewards~~ records.

* Left outer join : Returns all records from the left table and matched records from the right table. If there is no match the result is null.

Syntax:

Select table1.column, table2.column from table1 t1, table2 t2 where t1.column(+) = t2.column;

Eg:

Select ename, dname from employee e department d where e.dept not = d.dno;

Right outer join:- Returns all the records from the right table and matched records from the left table. If there is no match the result is null.

Syntax:-

Select table1.column, table2.column from tabel1 t1, table2 t2 where t1.column = t2.column (+);

Eg:

Select ename, dname from employee e, department d where e.dept no = d.dno (+);

3) Cross join:-

A cross join returns the cartesian product. This means that the join combines every row from the left table with every row from the right table.

Syntax:

Select column_name (s)
From Tabel 1
Cross Join table 2;

Eg:

ENO	ENAME	PNO	PNAME
E1001	Janani	P001	Soap
E1002	Jhanvi	P002	Shampoo

E1001	Janani	P001	Soap
E1001	Janani	P002	Shampoo
E1002	Jhanvi	P001	Soap
E1002	Jhanvi	P002	Shampoo

4) Self join :-

A Self join is one way the same table is involved in the join operation.

Syntax:

```
Select Coloumn.name (s)
From table_name As alias 1
JOIN table_name As alias 2
on alias.coloumn_name = alias.coloumn_
                                name;
```

Eg:

```
Select e1.name, e1.salary from employee e1,
employee e2 * where (e1.sal > e2.sal) and e2
ename = 'sita';
```

5) Full join :-

Full Join returns all records from both tables, including matching and non-matching records.

Syntax:

```
SELECT coloumn1, coloumn2
FROM tabel 1
FULL JOIN table 2
ON tabel.coloumn = tabel2.coloumn;
```

Eg:

```
Select Employee.Name,
Department.Department_Name
From Employee
Full Join Department
ON Employee.Department_ID =
Department.Department_ID;
```